Appl Intell (2015) 43:676–694
DOI 10.1007/s10489-015-0658-x

Evolutionary algorithms with user's preferences for solving hybrid interval multi-objective optimization problems

Dunwei Gong · Yiping Liu · Xinfang Ji · Jing Sun

Published online: 19 May 2015
© Springer Science+Business Media New York 2015

Abstract Hybrid interval multi-objective optimization problems are common in real-world applications. These problems involve both explicit and implicit objectives, and the values of these objectives are intervals. Few previous methods are suitable for them. An evolutionary algorithm with a large population and a user's interval preferences was presented to effectively solve the problems in this paper. In the proposed algorithm, a similarity-based strategy was employed to estimate the interval values of implicit objectives of evolutionary individuals that the user had not evaluated in order to alleviate user fatigue; the user's preferences to different objectives were expressed precisely as intervals by solving an auxiliary optimization problem; a sorting scheme based on the user's preferences was proposed to guide the population evolving toward the user's preferred regions. We applied the proposed method to an interior layout problem, which is a typical optimization problem with both interval parameters in the explicit objective and interval value of the implicit objective. The proposed algorithm was compared with four other optimization algorithms on the interior layout problem. Experimental results validated its effectiveness and superiority over the compared algorithms in terms of solution quality and the number of user's evaluations.

Keywords Evolutionary optimization · Hybrid objectives · Interval objectives · User preference

D. W. Gong · Y. P. Liu (✉)
School of Information and Electrical Engineering,
China University of Mining and Technology, Xuzhou, China
e-mail: lyp.cumt@gmail.com

D. W. Gong
e-mail: dwgong@vip.163.com

D. W. Gong
School of Electrical Engineering and Information Engineering,
LanZhou University of Technology, Lanzhou, China

X. F. Ji (✉)
Department of Mechanical, Power and Information Engineering,
China University of Mining and Technology, Yinchuan College,
Yinchuan, China
e-mail: mimosa_615615@126.com

J. Sun
School of Science, Huai Hai Institute of Technology,
Lianyungang, China
e-mail: jing8880@sina.com

1 Introduction

Many real-world optimization problems include both explicit (quantitative) and implicit (qualitative) objectives, of which explicit ones can be expressed with well-defined functions. Implicit objectives, however, can only be described by human languages, opinions or thoughts, and these descriptions vary from person to person, time to time and place to place. These problems are referred to as hybrid multi-objective optimization problems (MOPs) [1]. Taking the problem of fashion design as an example, the coordination of colors in different parts is an explicit objective. The user's evaluation of it, however, is an implicit one [2]. Sometimes it is difficult to determine the values of the uncertain parameters contained in explicit objectives. In addition, since implicit objectives are difficult to quantify, human evaluations on them are subjective, resulting in uncertain evaluation results. Such problems are called hybrid uncertain MOPs.

It is well known that uncertain parameters contained in optimization problems can be classified into the following three categories: stochastic variables, fuzzy numbers and intervals [3]. When an uncertain parameter is a stochastic variable, the probability distribution with which the stochastic variable obeys needs to be known. When it is a fuzzy number, the membership function that describes the fuzzy number needs to be given. However, it is difficult to obtain either accurate probability distribution or appropriate membership function in practical applications. When an uncertain parameter is an interval, its upper and lower limits or its midpoint and width are required to be known beforehand. Since the values of these parameters are easy to access, studies on optimization problems with interval parameters are relatively fruitful. We focused on hybrid interval MOPs (HIMOPs) in this study.

Various methods for solving both hybrid MOPs and interval MOPs exist. Nevertheless, few methods are suitable for HIMOPs. Since these problems have double complexities, i.e., hybridity and uncertainty, it is very difficult to solve them. To the best of our knowledge, there only exists our recently proposed method [4], which provides an effective candidate for HIMOPs. In that method, a HIMOP is transformed into a precise one, and an evolutionary algorithm is proposed to solve the transformed problem. It can be observed that transforming an interval into a precise value is rather subjective from the view of transformation methods, suggesting that it is better to directly tackle the interval objectives. A similarity-based estimation method is employed to estimate the precise values of implicit objectives of an evolutionary individual in order to alleviate user fatigue when a large population is adopted for evolution. It, however, is not suitable for estimating interval values. In addition, the proposed method aims at finding a well-converged and evenly-distributed Pareto optimal set, which is also the purpose of most state-of-the-art evolutionary multi-objective optimization algorithms. However, in practice, a user is often interested in only one solution. Thus, two tasks, i.e., optimization for finding Pareto optimal solutions and decision-making for choosing one satisfactory solution [5], need to be combined to obtain the most satisfactory solution for the user. The relationship between the two tasks leads to three different kinds of methods: the first is making decision before optimization, called *a priori* methods; the second is making decision after optimization, called *a posteriori* methods; and the last is making decision during optimization, called interactive methods. Undoubtedly, the user's preferences are prerequisite to make the decision. Since the implicit objectives of the problem considered in this study are evaluated by the user, it is very easy to interactively express his/her preferences.

Based on the above consideration, when tackling HIMOPs using evolutionary algorithms, there are several difficulties should be properly solved. First, the user's preference to different objectives should be adaptively changed during the evolution for guiding the population to an interesting region. Second, the implicit objectives of individuals should be automatically evaluated to alleviate the user fatigue when a large population is adopted. Last but not least, the individuals with interval objectives should be assigned a fitness value by incorporating the user's preference for improving the selection pressure.

Therefore, in this study, the user interactively expressed his/her preferences to different objectives, expressed with intervals and called interval preferences. Besides, in order to alleviate user fatigue and to accelerate the speed of searching the most satisfactory solution, a similarity-based strategy for evaluation and a large population were both employed by combining the sorting scheme based on the user's preferences in the framework of NSGA-II [6].

The primary contributions can be summarized as the following three aspects. First, for evolutionary individuals not evaluated by the user, a similarity-based strategy was presented to estimate their implicit objectives whose values are intervals, which can enlarge the population size. The performance of the algorithm in exploration is improved because much more solutions can be searched. Second, for the user's interval preferences to both the explicit and the implicit objectives, a precision method was given by solving another optimization problem, which makes the user's preference more easily being incorporated into the course of optimization. Finally, a sorting scheme based on the user's preferences was designed to guide the search toward the user's preferred region.

The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 proposes our method, which includes a similarity-based estimation of the interval values of implicit objectives, precision of the user's interval preferences, and a sorting scheme based on the user's preferences. The applications of the proposed method in the interior layout design problem, a typical HIMOP, and comparative experiments are given in Sections 4 and 5. Finally, Section 6 concludes this study and offers suggestions on possible opportunities for future research.

2 Related work

As mentioned in Section 1, the HIMOP has both implicit and interval objectives, and is considerably difficult to solve. In this section, the detailed definition of HIMOP is first given. Then, previous studies on hybrid MOPs and interval MOPs are respectively reviewed in the following two paragraphs. Finally, evolutionary optimization methods based on the users preference are introduced.

We considered the following HIMOP:

$$\begin{aligned} &\max f_1(x, b_1), \cdots, f_P(x, b_P), f_{P+1}(x), \cdots, f_{P+Q}(x) \\ &\text{s.t.} \quad x \in \mathrm{S} \subseteq R^n \end{aligned} \tag{1}$$

where x represents an n-dimensional decision variable belonging to domain S, $f_i(x, b_i)$, $i = 1, 2, \cdots, P$ is the i-th explicit objective with interval parameters, expressed with a well-defined function and calculated by a computer, b_i is a parameter described with an interval vector, $b_i = (b_{i1}, b_{i2}, \cdots, b_{iK})^T$, b_{ik} is the k-th component with $\underline{b}_{ik}$ and $\overline{b}_{ik}$ being its lower and upper limits, respectively, i.e., $b_{ik} = \left[\underline{b}_{ik}, \overline{b}_{ik}\right]$, $k = 1, 2, \cdots, K$, $i = 1, 2, \cdots, P$, and $f_{P+q}(x)$, $q = 1, 2, \cdots, Q$, $Q \geq 0$, is the q-th implicit objective expressed as an interval and given by a user.

As can be observed from formula (1), if P is equal to zero, formula (1) represents an optimization problem with implicit objectives; if Q is equal to zero, formula (1) represents an optimization problem with explicit objectives; if both P and Q are larger than zero, formula (1) represents an ordinary HIMOP.

Until now, there have been various studies on hybrid MOPs. In the study of Zhou et al. the values of various objectives obtained in different ways are scaled, the fitness of an evolutionary individual is calculated by weighing them, and the proposed algorithm was applied in fashion design [2]. In the method proposed by Kamalian et al. after a population has evolved for a number of generations using a traditional multi-objective genetic algorithm (GA) to optimize multiple explicit objectives, the user evaluates optimal solutions obtained in the performance of the implicit objectives, and offspring are produced based on the user's evaluations. Therefore, restrictions to both the population size and the number of generations resulting from the human-computer interface are not necessary, hence alleviating user fatigue. This method is also called sequential interactive genetic algorithm (IGA) [7]. In the method proposed by Brintrup et al. an optimization problem with several implicit objectives is focused on, and an evolutionary individual is evaluated in the performance of these implicit objectives combining IGAs with fuzzy systems. Having formulated the MOP together with the explicit objectives, they then solved it by using NSGA-II, and the proposed algorithm, called multi-objective IGA, was applied to an interior layout problem [8]. In the work of Brintrup et al., the performance of sequential IGA and multi-objective IGA is further compared, and verified that the latter is better than the former in solving a floor-planning problem [9]. The above methods can effectively solve hybrid MOPs. However, they are unsuitable for solving MOPs with intervals.

Interval multi-objective evolutionary algorithms are a kind of algorithms which directly solve interval MOPs. In the study of Limbourg et al., an evolutionary optimization method is utilized to solve the multi-objective optimization problem whose objectives contain noises, and the objective values are expressed with intervals. Two criteria, i.e., dominant relationship and hypervolume based on intervals, are used to measure the quality of a solution and the performance of a Pareto-optimal set, respectively [10]. In the method proposed by Li et al. an interval number programming method based on the order relation and the possibility degree of interval number is used to transform an uncertain optimization problem into the corresponding deterministic multi-objective optimization problem [11]. Unfortunately, the above methods are unsuitable for solving the MOPs involving implicit objectives.

These evolutionary optimization methods used to solve MOPs attempt to find their well-converged and evenly-distributed Pareto optimal sets. In practice, only one of the optimization tasks is to find an appropriate Pareto-optimal set. To get the most satisfactory solution for the user, another task, i.e., decision-making, is also very important. During evolutionary optimization, the sequence of these two tasks determines the following three methods of the evolution and decision-making, i.e., *a priori*, *a posteriori* and interactive methods [12]. For the *a priori* method, the user's preferences are first specified, and then incorporated into the process of optimization to find one (or several) solution(s) that meet the user's requirements. Although many selections of the user are not necessary at the end of the evolution, it is difficult for he/she to provide accurate preferences when his/her cognition on optimization problems is insufficient, resulting in optimal solutions with low quality. For the *a posteriori* method, a Pareto optimal solution set is first obtained by using appropriate approaches, and then submitted to the user for selection. Although the user's precise preferences can be obtained, a large number of optimal solutions may be generated, among which many are disliked by the user, increasing the user's burden in selecting the ones that he/she likes. For the third method, i.e., the interactive one, the user makes decisions during the evolution. This method is helpful not only to clarify the user's preferences, but also to reduce computation during optimization.

The interactive method has advantages over both the *a priori* and the *a posteriori* methods. The user's preferences in this method can be embedded in the beginning of the evolution, and evolved along with change of the user's cognition. In addition, one (or several) solution(s) meeting the user's requirements can be obtained after the evolution. In the study of Parmee et al., for piecewise objectives being optimized, the user is asked to specify which one is more important, and a co-evolutionary multi-objective genetic algorithm is employed to search for the optimal solution that satisfies the user [13]. In the work of Chaudhuri et al. the user's preferences are incorporated into the course of optimization, and the user concentrates only on one (or several) region(s) of interest after the whole Pareto front has been

obtained until the most satisfactory solution is found [14]. In the method proposed by Deb et al. a strictly increasing function is constructed to reflect the user's preferences after the population has evolved for a number of generations set in advance. Then the function is used to compare different solutions to guide the population evolving towards the most preferred region [15]. In the study of Sinha et al. a preference-based methodology is proposed, where the preferences provided by the user in the intermediate runs of an evolutionary multi-objective optimization algorithm is used to construct a polyhedral cone, which is used to eliminate a part of search space and conduct a more focused search. The dominance principle is modified to search for better solutions lying in the preferred region [16].

Expression of the user's preferences is also important besides the means of embedding them. There are various expression forms of the user's preferences, e.g., reference point [17], reference direction [18], solution ranking [19], and so on. In the work of Bently et al. the importance of objectives is defined, allowing evolutionary algorithms to converge on a small subset of acceptable non-dominated solutions [20]. It is convenient and common to adopt weights to represent importance values of different objectives. However, it is difficult and less reasonable to specify a precise weight for the user *a priori* because of the uncertainty of the user's cognition. Therefore, the user's preferences to different objectives were expressed with intervals in this study to alleviate the user's burden resulting from assigning precise weights. Note that both the optimization problem and the user's preference have uncertainties, these make the problem considered here more complex. Therefore, appropriate methods are required to make the user's preferences precise. In the work of Wang et al., a method of calculating the user's precise preferences is obtained by deduction, and different schemes are compared based on the corresponding intervals accumulated from all attributes [21]. In this study, the user's interval preferences were made precise by solving an auxiliary optimization problem in view of the idea of [21].

3 Proposed method

3.1 Overview

In the framework of NSGA-II, a novel evolutionary optimization algorithm with a user's preferences to solve the problem described with formula (1) is presented in this section. In the algorithm, the user is requested to input his/her interval preferences to different objectives before the evolution; a large population is adopted and the$K$-means clustering [25] is utilized to reasonably cluster it; then the computer is employed to calculate the values of explicit objectives of all evolutionary individuals, and the user evaluates only the values of implicit objectives of the centers of all clusters; for evolutionary individuals not evaluated by the user, a similarity-based estimation strategy is used to estimate the values of implicit objectives described in subsection 3.3. The user's interval preferences to different objectives are quantified by solving a single-objective optimization problem with constraints, elaborated in subsection 3.4. In addition, a sorting scheme based on the user's preferences is proposed to steer the search towards the user's preferred region, given in subsection 3.5. The user can modify his/her preferences during the evolution. The steps of the proposed method are as follows:

1. Set the values of evolutionary control parameters in the method, and initialize a population $x(t)$, $t = 0$.
2. Input the user's interval preferences to each objective.
3. Divide $x(t)$ into N_c clusters using the method in [25].
4. Decode each evolutionary individual, evaluate all implicit objectives of the centers of all clusters by the user, estimate the values of all implicit objectives of the rest evolutionary individuals in all clusters according to (4), and calculate the values of all explicit objectives of all evolutionary individuals by the computer.
5. Make the user's interval preferences precise by solving an auxiliary optimization problem.
6. Employ the tournament selection with size 2, and perform crossover and mutation operations to create offspring populations.
7. Combine parent and offspring populations.
8. Rank the combined population according to the sorting scheme based on the user's preferences, and select superior individuals to form a new population, $x(t+1)$.
9. Judge whether termination criteria have been met or not, if yes, stop the evolution and output the optimal solutions; otherwise, let $t = t + 1$. If the user's preferences are changed, go to Step 2; otherwise, go to Step 3.

The flow chart of the proposed method is depicted as Fig. 1 according to the above steps, and the shaded parts will be introduced in the following subsection.

3.2 Relation to our previous work

As stated in Section 1, based on the characteristics of uncertainties contained in the two types of objectives, HIMOP is first transformed into one without uncertainties by using interval analysis, and then an evolutionary optimization algorithm is employed to effectively solve the transformed problem by combining a traditional GA [4]. To be solved, however, is the optimization problem without uncertainties, transformed from the original one. In our recently proposed method for interval MOPs, a novel interval

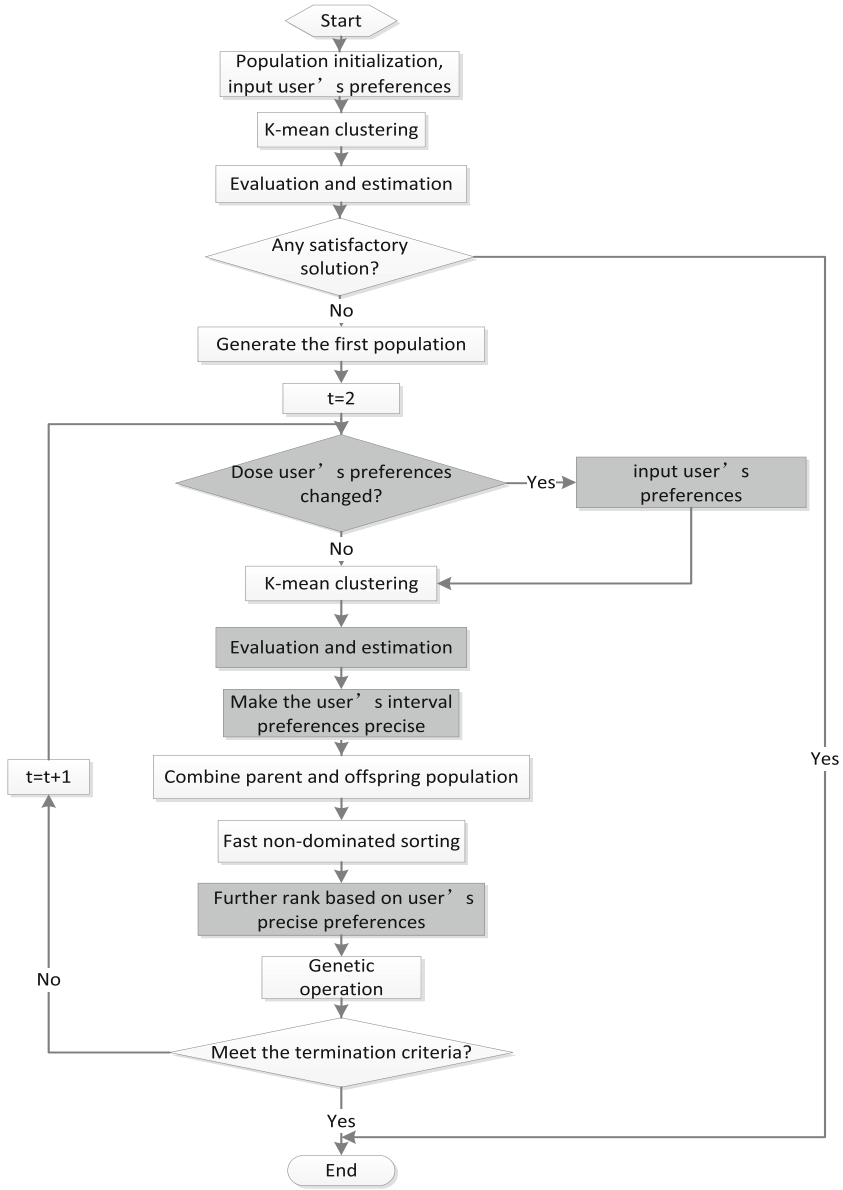


Fig. 1 Flow chart of the proposed method

dominance relation is defined to compare different individuals, and a crowding distance based on intervals is presented to further distinguish the individuals with the same rank [22]. In this study, HIMOP is directly solved by using the proposed method in [22].

As well known, when solving optimization problems with implicit objectives, a small population is often adopted due to the restriction of the human-computer interface, impacting on the capability of an algorithm in exploration. Adopting a large population can improve the algorithms capability in exploration. However, it also considerably increases the user fatigue since regular methods ask the user to evaluate objectives of every individual. To overcome this drawback, we presented a large population IGA [23]. In this algorithm, the population is divided into several clusters, and the user only evaluates the centers of all clusters. For individuals not evaluated by the user, a similarity-based strategy is employed to estimate their fitness. Moreover, a large population evolutionary algorithm to solve hybrid MOPs was proposed [24]. In this algorithm, the number of individuals evaluated by the user varies from generation to generation. For individuals not evaluated by the user, a similarity-based strategy is employed to estimate their fitness after the whole population has been divided into several clusters by using K-mean clustering. The above algorithm extends the search scope of the population, and has a good performance in exploration on condition of not increasing the user fatigue. Motivated by them, we adopt a large population in this study, and extend the above methods to estimate the interval values of implicit objectives.

3.3 Estimation of interval values of implicit objectives

In view of previous studies, the following techniques are still to be addressed to fulfill the proposed method: the first is the estimation of the interval values of implicit objectives; the second is the precision of the user's interval preferences to different objectives, and the last is the comparison of different evolutionary individuals based on the precise preferences. In this subsection, we focus on the first one.

As mentioned before, the values of explicit objectives can be calculated by a computer, and those of implicit ones, however, are to be evaluated by user, who can only evaluate a few evolutionary individuals due to user fatigue. To enlarge the population size, a similarity-based strategy was present to estimate the interval values of implicit objectives of evolutionary individuals that are not evaluated by the user in this subsection. First, the population is clustered according to the similarity of evolutionary individuals; thereafter, the user evaluates the implicit objectives of the center of each cluster; finally, the implicit objectives of the evolutionary individuals not evaluated by the user are estimated, which is the focus of this subsection. The phenotypes (values of implicit objectives) of an individual are determined mainly by its genotype in addition to some other factors. Generally, individuals with similar genotypes will have similar phenotypes. Therefore, it is reasonable to estimate the implicit objective values of the unevaluated individuals based on the evaluated ones with similar genotype.

Denote the population in the t-th generation as $x(t)$, its i-th evolutionary individual as $x_i(t)$ $(i = 1, 2, \cdots, N)$, whose genotype is $x_i(t) = [x_{i1}x_{i2}\cdots x_{iM}]^T$, where N and M are the population size and the number of variable, respectively. For two evolutionary individuals $x_j(t)$ and $x_i(t)$, their similarity, denoted as $\alpha(x_j(t), x_i(t))$, can be obtained by (2) [23], and its range is [0,1].

$$\alpha_m(x_j(t), x_i(t)) = \begin{cases} 1 & if \quad x_{jm} = x_{im} \\ 0 & if \quad x_{jm} \neq x_{im} \end{cases}$$

$$\alpha(x_j(t), x_i(t)) = \frac{1}{M}\sum_{m=1}^{M}\alpha_m(x_j(t), x_i(t)) \tag{2}$$

In order to cluster $x(t)$, the number of clusters, denoted as N_c, is first determined according to the human-computer interface and the process of the evolution; then N_c clusters of $x(t)$, denoted as $\{c_1(t)\}, \{c_2(t)\}, \cdots, \{c_{N_c}(t)\}$, are obtained according to the similarity of evolutionary individuals using the method of [25]. Correspondingly, their centroids are $c_1(t), c_2(t), \cdots, c_{N_c}(t)$, respectively.

For cluster $\{c_i(t)\}$, the user only evaluates the implicit objectives of its center $c_i(t)$, so the values of $f_{P+1}(c_i(t)), f_{P+2}(c_i(t)), \cdots, f_{P+Q}(c_i(t))$ can be obtained based on the user's evaluations, and expressed with intervals. For other evolutionary individuals belonging to $\{c_i(t)\}$, e.g., $x_j(t)$, $x_j(t) \neq c_i(t)$, there are yet no values of their implicit objectives. So it is necessary to estimate these values using appropriate strategies to compare different evolutionary individuals.

For the case that user's evaluations on $c_i(t)$ are expressed with precise values, the following estimate of the q-th implicit objective of $x_j(t)$ was given in [24]:

$$\hat{f}_{P+q}(x_j(t)) = f_{P+q}(c_i(t))e^{-(1-\alpha(x_j(t),c_i(t)))} \tag{3}$$

As can be observed, the value of $\hat{f}_{P+q}(x_j(t))$ is precise as a result of the precise value of $f_{P+q}(c_i(t))$. For the HIMOP investigated in this study, however, the values of implicit objectives of an evolutionary individual are represented using intervals. So (3) is unsuitable for estimating the interval values. To this end, the following strategy of estimating is presented.

Denote user's evaluation on the q-th implicit objective of $c_i(t)$ as $[\underline{f}_{P+q}(c_i(t)), \overline{f}_{P+q}(c_i(t))]$. Based on the above interval, the estimate of the q-th implicit objective of $x_j(t)$, also an interval, can be obtained, and denoted as $[\underline{\hat{f}}_{P+q}(x_j(t)), \overline{\hat{f}}_{P+q}(x_j(t))]$. The interval is expected to have the following characteristics: the more $x_j(t)$ is similar to $c_i(t)$, the closer $[\underline{\hat{f}}_{P+q}(x_j(t)), \overline{\hat{f}}_{P+q}(x_j(t))]$ is to $[\underline{f}_{P+q}(c_i(t)), \overline{f}_{P+q}(c_i(t))]$, and the smaller the uncertainty degree of $[\underline{\hat{f}}_{P+q}(x_j(t)), \overline{\hat{f}}_{P+q}(x_j(t))]$ is (but no less than that of $c_i(t)$), suggesting more precision of the estimate. Especially, when $x_j(t)$ is equal to $c_i(t)$, $[\underline{\hat{f}}_{P+q}(x_j(t)), \overline{\hat{f}}_{P+q}(x_j(t))]$ is the same as $[\underline{f}_{P+q}(c_i(t)), \overline{f}_{P+q}(c_i(t))]$. In view of this, the estimate of the q-th implicit objective of $x_j(t)$ can be expressed as follows:

$$\begin{aligned} \underline{\hat{f}}_{P+q}(x_j(t)) &= \underline{f}_{P+q}(c_i(t)) - \delta(\alpha(c_i(t), x_j(t))) \\ \overline{\hat{f}}_{P+q}(x_j(t)) &= \overline{f}_{P+q}(c_i(t)) + \delta(\alpha(c_i(t), x_j(t))) \end{aligned} \tag{4}$$

where

$$\begin{aligned} \delta(\alpha(c_i(t), x_j(t))) = \lambda(1 - \alpha(c_i(t), x_j(t)))e^{-\alpha(c_i(t),x_j(t))} \\ \times(f_{P+q}^{\max} - f_{P+q}^{\min}) \end{aligned} \tag{5}$$

where λ is a positive set by the user, and equal to 0.1 in our experiments. $f_{P+q}^{\min}$ and $f_{P+q}^{\max}$ are the minimal and the maximal values of the q-th implicit objective, respectively, also set by the user.

From equations (4) and (5), we have the following observations: (1) the value of $\delta(\alpha(c_i(t), x_j(t)))$ decreases along with the increase of $\alpha(c_i(t), x_j(t))$, suggesting that the more similar $x_j(t)$ is to $c_i(t)$, and the smaller the uncertainty degree of $[\underline{\hat{f}}_{P+q}(x_j(t)), \overline{\hat{f}}_{P+q}(x_j(t))]$; moreover, the above uncertainty degree is larger than or equal to that of $[\underline{f}_{P+q}(c_i(t)), \overline{f}_{P+q}(c_i(t))]$ due to $\delta(\alpha(c_i(t), x_j(t)))$ being

larger than or equal to zero, which is as expected and rational; (2) $\delta(\alpha(c_i(t), x_j(t)))$ is in the range of $[0, \lambda(f_{p+q}^{\max} - f_{p+q}^{\min})]$, suggesting that equation (4) accords with the aforementioned requirements; (3) $[\hat{\underline{f}}_{P+q}(x_j(t)), \hat{\overline{f}}_{P+q}(x_j(t))]$ has the same midpoint as $[\underline{f}_{P+q}(c_i(t)), \overline{f}_{P+q}(c_i(t))]$, indicating that for the q-th implicit objective, $x_j(t)$ has the same expectation as $c_i(t)$. The reason is that the user has limited capability in distinguishing different evolutionary individuals that belong to the same cluster.

3.4 Precision of user's interval preferences to different objectives

The user's preferences to different objectives, including both the explicit and the implicit ones, are represented with intervals to reflect the uncertainty of the user's cognition. Before incorporating them into the course of optimization, a precision method of the user's interval preferences was proposed in this subsection.

Denote the user's preference to the j-th objective as $[\underline{w}_j, \overline{w}_j]$. The problem to be solved in this subsection is to select a precise value, denoted as w_j, from $[\underline{w}_j, \overline{w}_j]$ using appropriate methods to optimize an objective function set in advance. The precise value is called the user's precise preference to the j-th objective, and the above problem is essentially an optimization problem.

For the optimization problem represented with formula (1), denote the archive after population $x(t)$ has evolved for t generations as $A(t)$. It is the Pareto-optimal set up to the present, and contains $|A(t)|$ optimal solutions. The i-th optimal solution of $A(t)$ is denoted as $a_i(t)$, $i = 1, 2, \cdots, |A(t)|$ and the value of its j-th objective as $[\underline{f}_j(a_i(t)), \overline{f}_j(a_i(t))]$. Different optimal solutions have different values to the j-th objective. For all optimal solutions in $A(t)$, the minimal lower limit and the maximal upper limit of these values, denoted as $\underline{f}_j^{\min}(A(t))$ and $\overline{f}_j^{\max}(A(t))$, respectively, can be expressed as follows:

$$\begin{aligned} \underline{f}_j^{\min}(A(t)) &= \min_{i=1,2,\cdots,|A(t)|} \underline{f}_j(a_i(t)) \\ \overline{f}_j^{\max}(A(t)) &= \max_{i=1,2,\cdots,|A(t)|} \overline{f}_j(a_i(t)) \end{aligned} \tag{6}$$

Denote the normalized value of the j-th objective of $a_i(t)$ as $[\underline{f}'_j(a_i(t)), \overline{f'_j}(a_i(t))]$. The relation between $[\underline{f}'_j(a_i(t)), \overline{f'_j}(a_i(t))]$ and $[\underline{f}_j(a_i(t)), \overline{f_j}(a_i(t))]$ is as follows:

$$\begin{aligned} \underline{f}'_j(a_i(t)) &= \frac{\underline{f}_j(a_i(t)) - \underline{f}_j^{\min}(A(t))}{\overline{f}_j^{\max}(A(t)) - \underline{f}_j^{\min}(A(t))} \\ \overline{f'_j}(a_i(t)) &= \frac{\overline{f}_j(a_i(t)) - \underline{f}_j^{\min}(A(t))}{\overline{f}_j^{\max}(A(t)) - \underline{f}_j^{\min}(A(t))} \end{aligned} \tag{7}$$

It is expected that the weighted sum of all midpoints of all objectives of optimal solutions in $A(t)$ is maximal by selecting the value of w_j from $[\underline{w}_j, \overline{w}_j]$, indicating that an appropriate value of w_j will make the accumulation of all objectives of optimal solutions in $A(t)$ optimal. Since $A(t)$ has a relation with the number of generations, t, w_j should be a function associated with t, and expressed with $w_j(t)$, indicating that different values of $w_j(t)$ can be selected from $[\underline{w}_j, \overline{w}_j]$ in different generations. Therefore, the problem of making the user's interval preferences precise can be transformed into the following single-objective optimization problem with constraints [21]:

$$\begin{aligned} &\max \textstyle\sum_{i=1}^{|A(t)|} \sum_{j=1}^{P+Q} \mathrm{mid}([\underline{f}'_j(a_i(t)), \overline{f'_j}(a_i(t))]) w_j(t) \\ &\text{s.t. } \underline{w}_j \le w_j(t) \le \overline{w}_j \\ &\quad \textstyle\sum_{j=1}^{P+Q} w_j(t) = 1 \end{aligned} \tag{8}$$

where $\mathrm{mid}([\underline{f}'_j(a_i(t)), \overline{f'_j}(a_i(t))]) = \frac{\underline{f}'_j(a_i(t)) + \overline{f'_j}(a_i(t))}{2}$. A group of user's precise preferences, i.e., precise weights of different objectives, $w_1(t), w_2(t), \cdots, w_{P+Q}(t)$, can be obtained by solving the maximization problem above.

As can be observed from the process above, precision of the user's interval preferences are obtained at each generation. Therefore, the precise weights change along with the evolution within the given preference intervals to optimize the objective in formula (8), suggesting that interval preferences are flexible. In addition, it is unnecessary to specify precise weights in advance, which can alleviate the user's burden resulting from assigning precise weights.

For $a_i(t)$, the weighted sum of all values of its objectives using weights $w_1(t), w_2(t), \cdots, w_{P+Q}(t)$ is still an interval, and expressed as $\left[\sum_{j=1}^{P+Q} \underline{f}'_j(a_i(t)) w_j(t), \sum_{j=1}^{P+Q} \overline{f'_j}(a_i(t)) w_j(t)\right]$.

3.5 Sorting scheme based on user's preferences

We incorporated the user's preferences into the sorting scheme of evolutionary individuals in this subsection. When adopting interval dominance relationship [22] to sort evolutionary individuals, they are grouped into several categories with different ranks assigned. Evolutionary individuals with the same rank are non-dominated, therefore, other indicators, e.g., crowding distance [6], are necessary to distinguish them. The user's preferences were used in this study. To achieve that, $P + Q$ interval objective values can be transformed into one based on the precise weights obtained in subsection 3.3; thereafter, evolutionary individuals with the same rank can be further distinguished by using an interval order. For more details, please refer to [26]. Of course, the weighted sum interval can be further transformed into a precise value by adopting appropriate methods, and then

these evolutionary individuals can be further distinguished by comparing the transformed precise values. Since the comparison of intervals is more difficult than that of precise values, the latter was adopted in this study.

The following tackles the problem of how to transform the weighted sum interval of an evolutionary individual into a precise value, called the user's satisfactory degree to the evolutionary individual, to further distinguish evolutionary individuals with the same rank. To reflect the user's satisfactory degree to an evolutionary individual, two reference hyper-cuboids, called the superior and the inferior hyper-cuboids, in the objective space are set. Correspondingly, two intervals, called the superior and the inferior intervals, respectively, can be obtained by weighting all objectives of the superior hyper-cuboid and those of the inferior hyper-cuboid. For an evolutionary individual, the user's satisfactory degree to it can be calculated by the distances from the weighted sum interval of the evolutionary individual to the two intervals.

Considering the maximization problem, the superior and the inferior hyper-cuboids are composed of the maximal and minimal values of the objectives of optimal solutions in $A(t)$, respectively. To be more specific, the maximal lower limit and the minimal upper limit of the j-th objective, denoted as $\underline{f}_j^{\max}(A(t))$ and $\overline{f}_j^{\min}(A(t))$, respectively, can be derived as follows:

$$\begin{aligned} \underline{f}_j^{\max}(A(t)) &= \max_{i=1,2,\cdots,|A(t)|} \underline{f}_j(a_i(t)) \\ \overline{f}_j^{\min}(A(t)) &= \min_{i=1,2,\cdots,|A(t)|} \overline{f}_j(a_i(t)) \end{aligned} \tag{9}$$

Combining equation (6), we can see that $\prod_{j=1}^{P+Q} \otimes \left[\underline{f}_j^{\max}(A(t)), \overline{f}_j^{\max}(A(t))\right]$ and $\prod_{j=1}^{P+Q} \otimes \left[\underline{f}_j^{\min}(A(t)), \overline{f}_j^{\min}(A(t))\right]$ are the superior and the inferior hyper-cuboids, respectively.

Based on the user's precise preferences $w_1(t)$, $w_2(t), \cdots, w_{P+Q}(t)$, the superior and the inferior intervals, denoted as $I_{\mathrm{s}}(t)$ and $I_{\mathrm{i}}(t)$, respectively, can be obtained as follows:

$$\begin{aligned} I_{\mathrm{s}}(t) &= [\sum_{j=1}^{P+Q} \underline{f}_j'^{\max}(A(t))w_j(t), 1] \\ I_{\mathrm{i}}(t) &= [0, \sum_{j=1}^{P+Q} \overline{f}_j'^{\min}(A(t))w_j(t)] \end{aligned} \tag{10}$$

where

$$\begin{aligned} \underline{f}_j'^{\max}(A(t)) &= \frac{\underline{f}_j^{\max}(A(t)) - \underline{f}_j^{\min}(A(t))}{\overline{f}_j^{\max}(A(t)) - \underline{f}_j^{\min}(A(t))} \\ \overline{f}_j'^{\min}(A(t)) &= \frac{\overline{f}_j^{\min}(A(t)) - \underline{f}_j^{\min}(A(t))}{\overline{f}_j^{\max}(A(t)) - \underline{f}_j^{\min}(A(t))} \end{aligned} \tag{11}$$

Taking the bi-objective maximization problem as an example, as shown in Fig. 2, after the population has evolved for several generations, the hyper-cuboids corresponding to the three Pareto optimal solutions in $|A(t)|$ are depicted with A, B and C. In addition, D and E are the superior and the inferior hyper-cuboids, respectively.

For the i-th evolutionary individual, $x_i(t)$, of $x(t)$, denote the weighted sum interval of all objectives of $x_i(t)$ as $I(x_i(t))$, i.e.:

$$I(x_i(t)) = \left[\sum_{j=1}^{P+Q} \underline{f}_j'(x_i(t))w_j(t), \sum_{j=1}^{P+Q} \overline{f}_j'(x_i(t))w_j(t)\right] \tag{12}$$

where $[\underline{f}_j'(x_i(t)), \overline{f}_j'(x_i(t))]$ are the normalized value of j-th objective of $x_i(t)$.

The distance between $I(x_i(t))$ and $I_{\mathrm{s}}(t)$ can be calculated by using the method in [27], denoted as $d(I(x_i(t)), I_{\mathrm{s}}(t))$. Similarly, the distance between $I(x_i(t))$ and $I_{\mathrm{i}}(t)$, denoted as $d(I(x_i(t)), I_{\mathrm{i}}(t))$, can also be obtained. Then the user's satisfactory degree to $x_i(t)$, denoted as $v(x_i(t))$, can be measured as follows:

$$v(x_i(t)) = \frac{d(I(x_i(t)), I_{\mathrm{i}}(t))}{d(I(x_i(t)), I_{\mathrm{s}}(t)) + d(I(x_i(t)), I_{\mathrm{i}}(t))} \tag{13}$$

As can be observed from equation (13), the value of $v(x_i(t))$ is in the range of [0, 1]. The nearer $I(x_i(t))$ is to the superior interval as well as the farther from the inferior one, the larger is the value of $v(x_i(t))$. Accordingly, the better is $x_i(t)$. In particular, when $I(x_i(t))$ is equal to $I_{\mathrm{s}}(t)$, $v(x_i(t))$ has the maximal value of one, thereby, $x_i(t)$ is the best evolutionary individual. In contrast, when $I(x_i(t))$ is equal to

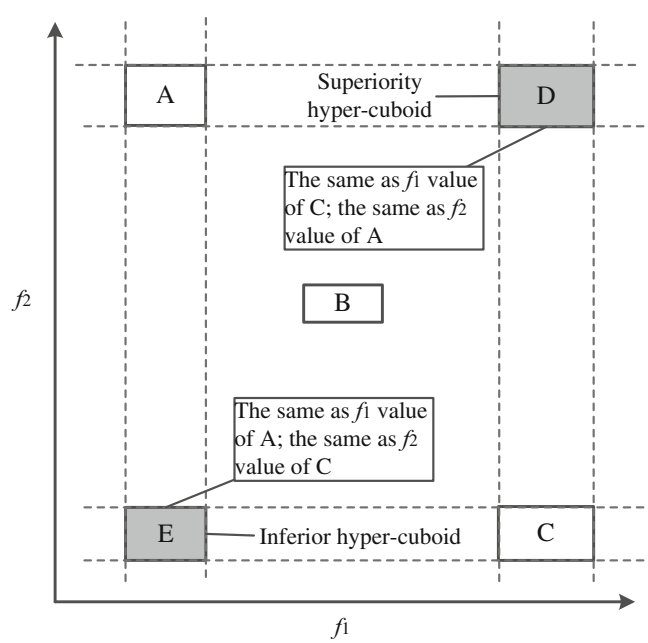


Fig. 2 Superior and inferior hyper-cuboids

$I_i(t)$, $v(x_i(t))$ has the minimal value of zero. Consequently, $x_i(t)$ is the worst evolutionary individual. Additionally, it is easy to know that if $x_A(t)$ dominates $x_B(t)$, $v(x_A(t))$ must be larger than $v(x_B(t))$. This means that this preference relationship preserves the existing dominance relationship and the search would not be biased towards undesired regions of the search space [12].

Based on the above discussion, the detailed sorting scheme is as follows: the interval dominance relationship proposed in [22] is adopted as the primary selection criteria to sort the evolutionary individuals; then those individuals with the same rank are further distinguished based on the user's satisfactory degrees to them. The smaller the rank of an evolutionary individual and the larger the user's satisfactory degree to it, the better its performance. The above method also provides an effective criterion for selecting appropriate evolutionary individuals as the objects of genetic operations.

4 Application to an interior layout problem

4.1 Interior layout problem

Consider the interior layout design of a general flat, as shown in Fig. 3 (reproduced from [4]). The goal of the optimization problem is to assign each part an appropriate width and length to minimize the total cost of the flat and to meet the user's aesthetic requirements. This is a typical hybrid MOP. To be more specific, it contains one explicit objective and one implicit objective. The explicit objective is the total cost of the flat, i.e., the sum of all parts' cost. The cost per unit area of each part, influenced by the relation between market supply and demand, is not a precise value, but fluctuates in a range, resulting in the total cost of the flat, denoted as $f_1(x, c)$, being an objective with parameters whose values are intervals. The implicit objective, denoted as $f_2(x)$, is to meet the user's aesthetic requirements. $f_2(x)$ cannot be expressed as a well-defined function and the user's cognition on the evaluated objects has uncertainties. If an interval is adopted to express the uncertainties above, the value of $f_2(x)$ is also an interval.

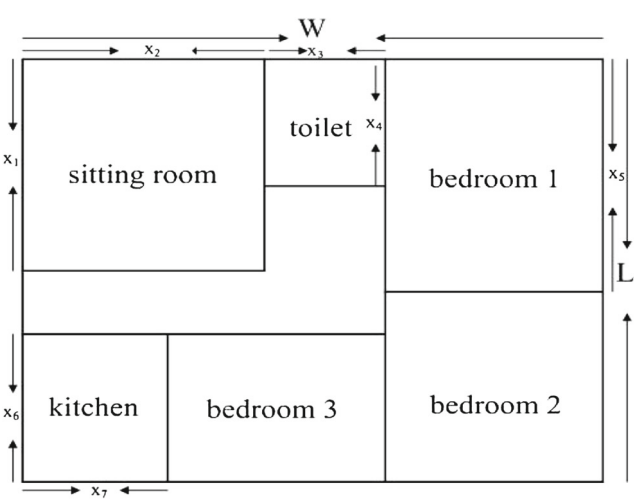


Fig. 3 Interior layout of flat

The values of the cost per unit area of different parts are different because of their different functions. Denote the cost per unit area of the sitting room, toilet, bedroom 1, kitchen, bedroom 3, bedroom 2, and aisle as $c_1, c_2, \ldots, c_7$, respectively, $c_k = [\underline{c_k}, \bar{c}_k]$, $k = 1, 2, \cdots, 7$, $c = (c_1, c_2, \cdots, c_7)^{\mathrm{T}}$, $x = (x_1, x_2, \ldots, x_7)^{\mathrm{T}}$ is decision variable, the above problem can be formulated as the following bi-objective optimization problem with intervals:

$$
\begin{aligned}
\min\ f_1(x, c) \quad &= c_1x_1x_2 + c_2x_3x_4 + c_3(W - x_2 - x_3)x_5 \\
&\quad + c_4x_6x_7 + c_5x_6(x_2 + x_3 - x_7) \\
&\quad + c_6(W - x_2 - x_3)(L - x_5) + c_7((x_2 + x_3) \\
&\quad \times (L - x_1 - x_6) + (x_1 - x_4)x_3) \\
\max;\ f_2(x) & \\
s.t.\quad & x_1 \in \{4.0, 4.3, 4.6, 4.9, 5.2\} \\
& x_2 \in \{4.0, 4.3, 4.6, 4.9, 5.2, 5.5, 5.8, 6.1, 6.4, 6.7, 7.0, 7.3\} \\
& x_3 \in \{2.0, 2.3, 2.6, 2.9, 3.2\} \\
& x_4 \in \{2.0, 2.4, 2.8, 3.2, 3.6\} \\
& x_5 \in \{1.0, 2.0, 3.0, 4.0, 5.0\} \\
& x_6 \in \{2.6, 2.9, 3.2, 3.5, 3.8\} \\
& x_7 \in \{1.0, 2.0, 3.0, 4.0, 5.0\}
\end{aligned} \tag{14}
$$

The value of each decision variable of the problem is chosen from the corresponding set composed of several discrete values, since the values of the width and the length of each part are often discrete in architectural standards. It can be seen from equation (14), the number of alternatives for decision variables $x_1, x_2, x_3, x_4, x_5, x_6$ and x_7 are 5, 12, 5, 5, 5, 5 and 5, respectively, the number of all possible interior layout designs is $12 \times 5^6 = 187500$.

4.2 Parameter settings

In our application, the values of width, W, and length, L, of the flat were equal to 12.5m and 10m, respectively. The value of the cost per unit area of each part is an interval, and that of c_1 was between 800 and 900, those of both c_2 and c_4 between 900 and 1100, those of c_3, c_5 and c_6 between 600 and 700, and that of c_7 between 400 and 600, respectively. To improve the capability of the algorithm in exploration, the population size, N, was set 200. Since the human-computer interface is limited, the largest number of evolutionary individuals displayed through the human-computer interface, $N_{\max}$, was equal to 12. The value of each decision variable was chosen from the corresponding

set composed of several discrete values, hence an allele of an evolutionary individual was one of those discrete values, suggesting that all genetic operators suitable for binary encoding, e.g., two-point crossover, intermediate recombination and uniform mutation, can be used in the proposed method. The experiments aimed at validating the effectiveness of the proposed method, rather than investigating the influence of genetic operators on the performance of the proposed method. As a consequence, only two simple genetic operators, i.e., one-point crossover and one-point mutation, were adopted, which will not affect the accuracy of experimental results. The probabilities of crossover and mutation were 0.95 and 0.01, respectively. In addition, taking the user's fatigue into consideration, the largest number of generations, T, was equal to ten, if it was not specified. In the experiments, 20 volunteers from our laboratory were invited as the users. Since the problem considered in the experiments contains two objectives only, i.e. f_1 and f_2, once the range of w_1 is assigned by the user, that of w_2 can be directly calculated by equation $w_2 = 1 - w_1$. Table 1 shows the users' interval preferences on f_1. However, if the user does not want to input any preference information, the default preferences, $W = (\frac{1}{P+Q}, \cdots, \frac{1}{P+Q})$, would be adopted. Since the optimization problem in the experiments has only two objectives, i.e., $P + Q = 2$, W=(0.5,0.5).

4.3 Human-computer interface

The interior layout design system was developed using Visual Basic 6.0 on the platform of Windows XP. Its human-computer interface, as shown in Fig. 4, includes the following four parts: (1) a sample of the interior layout in which the names of its all parts are shown; (2) parameter settings and statistical information about the evolution, including the current generation, the number of evolutionary individuals evaluated by the user and his/her time consumption, and the probabilities of crossover and mutation; (3) the phenotypes of the centers of all clusters and the values of their objectives, where the value of the explicit objective reflects an evolutionary individual's total cost. The smaller the value, the lower is its total cost. The upper and the lower limits assigned by the user were chosen from the integer set whose elements are in the range of [0, 100], and the upper limit of an evolutionary individual is not smaller than its lower one. The more satisfied the user is with an evolutionary individual, the larger are the upper and the lower limits; (4) command buttons for a population evolving, e.g., "Initialize", "Next Generation", and "End".

Table 1 Users' interval Preferences on f_1

User	w_1	User	w_1	User	w_1	User	w_1
1	[0.3, 0.7]	6	[0.0, 0.2]	11	[0.2, 0.7]	16	[0.5, 0.6]
2	[0.3, 0.7]	7	[0.4, 0.7]	12	[0.3, 0.6]	17	[0.2, 0.3]
3	[0.8, 1.0]	8	[0.1, 0.6]	13	[0.8, 1.0]	18	[0.7, 0.9]
4	[0.6, 0.9]	9	[0.6, 0.8]	14	[0.4, 0.7]	19	[0.2, 0.4]
5	[0.6, 0.7]	10	[0.0, 0.1]	15	[0.5, 1.0]	20	[0.4, 0.6]

4.4 Experiments on the behavior of the proposed method

In this subsection, the experiments were divided into three parts. The first was used to reveal the law of the users cognition w.r.t. the evolution; the second was done with the purpose of showing the course of precision of the users interval preferences; the goal of the third is to investigate the similarity-based strategy for estimating the interval values of implicit objectives.

4.4.1 User's cognition w.r.t. evolution

The user can continuously gather his/her experience during the interaction. Therefore, the more solutions the user evaluates, the more explicit the users preference is. To find the law of the user's cognition w.r.t. the evolution, 20 independent runs of the proposed method were performed. The interval values of the implicit objective of all centers assigned by the user in each generation were recorded, their mid-points and uncertainty degrees were automatically calculated by the system, and their averages were then obtained. Figure 5 shows the curves of the average mid-point and the uncertainty degree along with the evolution.

We can observe from Fig. 5 that, (1) in the initial phase of the evolution, the average mid-point of the implicit objective of all centers is small, and the average uncertainty degree is large, indicating that the performance of all centers is not good, and the user's cognition on the evaluated objects is inadequate; and (2) along with the evolution, the average mid-point of the implicit objective of all centers gradually increases and the average uncertainty degree gradually decreases, suggesting that the evolutionary individuals found by the system are better and the user's cognition on the evaluated objects is clearer. Therefore, adopting intervals to represent the evaluations on the implicit objective of evolutionary individuals can well reflect the user's cognition law.

4.4.2 Precision of user's interval preferences

In this subsection, users 1 and 2 were involved and they made the lower limit of the range increasing along with the evolution. The curve of the precision value of the user's preferences to the explicit objective w.r.t. the evolution was depicted in Fig. 6. Since the user's preferences to the implicit objective are opposite to Fig. 6, it is omitted for brevity.

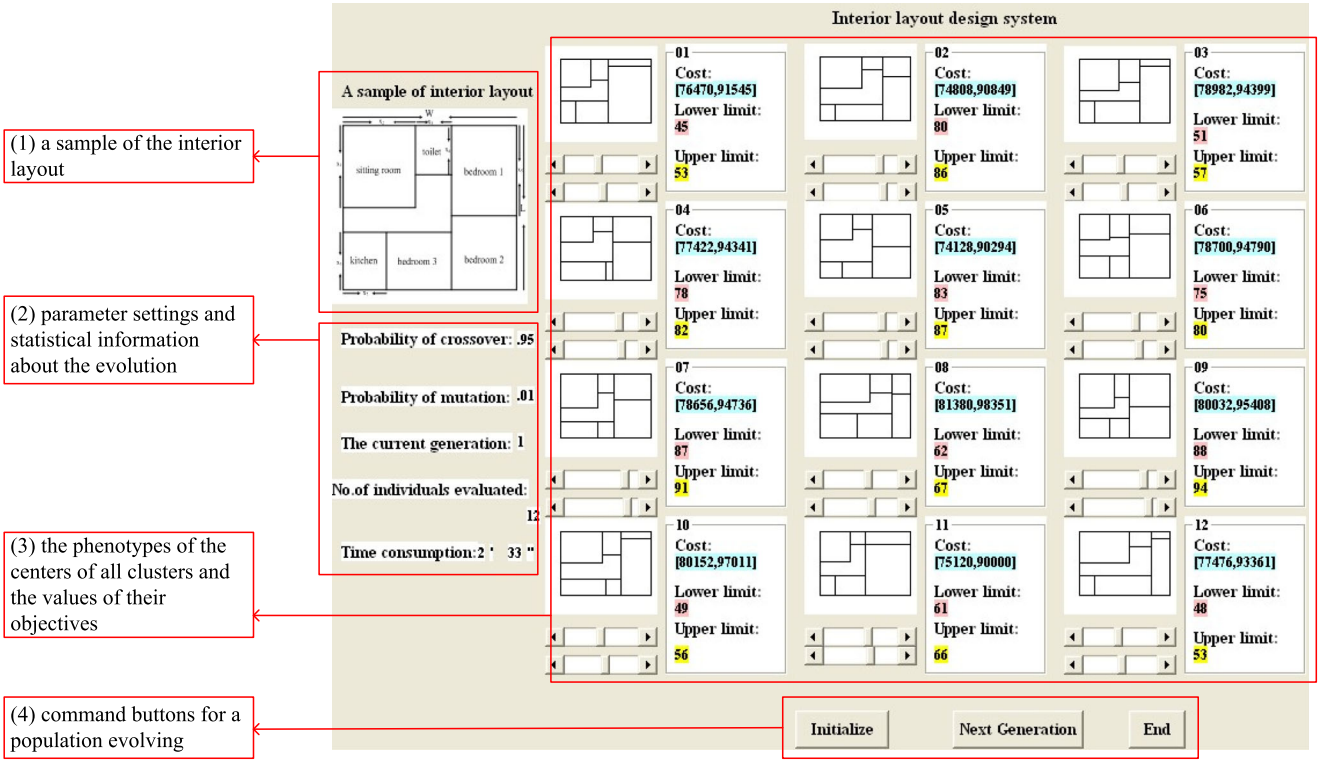


Fig. 4 Human-computer interface

It is evident from Fig. 6 that although the user's preferences to the explicit objective is in the range of [0.3, 0.7], its precise value changes along with the evolution, i.e., the user's precise preference is not constant during optimization. This change trend has a close relation with the values of the objective of the evolutionary individuals in $A(t)$ and the user's preferences.

4.4.3 Estimation the interval values of implicit objectives

In subsection 3.3, a similarity-based estimation strategy was adopted to estimate the interval values of implicit objectives. Since the strategy calculates the approximations of the real objective values, its error term was investigated in this subsection. 10 sets were randomly generated with each

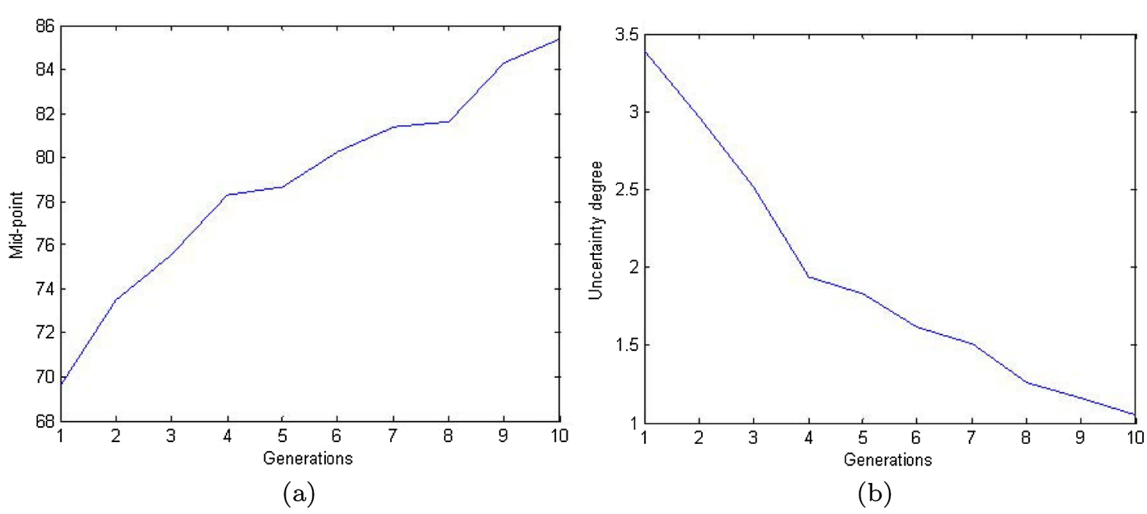


Fig. 5 Curves of average mid-point and uncertainty degree w.r.t. evolution

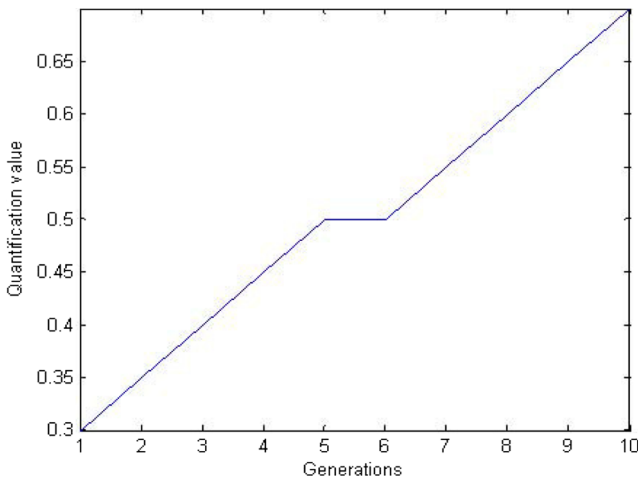


Fig. 6 Quantification value of user's preferences to explicit objective w.r.t. evolution

containing 200 solutions and being divided into 12 clusters. All the solutions in each set were evaluated by one user (user 1-10), whereas solutions that are not the cluster centers were also estimated by the similarity-based estimation strategy. The error term was expressed as follows:

$$\varepsilon = \frac{1}{N_{nc}} \sum_{i=1}^{N_{nc}} \frac{d(\hat{f}_2(x_i), f_2(x_i))}{f_2{}^{\max} - f_2{}^{\min}} \tag{15}$$

where

$$d(\hat{f}_2(x_i), f_2(x_i)) = \max\left\{|\underline{\hat{f}_2}(x_i) - \underline{f_2}(x_i)|, |\overline{\hat{f}_2}(x_i) - \overline{f_2}(x_i)|\right\} \tag{16}$$

where N_{nc} is the number of solutions that are not the cluster centers, and is equal to 188 in this study. $f_2{}^{\max}$ and $f_2{}^{\min}$ are the maximal and the minimal values of f_2, and are equal to 100 and 0, respectively. It is easy to know that the error term ranges from 0 to 1, and the smaller value it has, the more accurately the similarity-based estimation strategy estimates the solutions. Table 2 shows the error terms of these 10 sets.

From Table 2, the mean value of these error terms is 0.22. This means that the approximate values can reflect almost 80% real values, and the similarity-based estimation strategy has a good ability in estimating the interval objective values. Solutions with high approximate values will be selected for further optimization, and have a great potential in helping the user to search for his/her favorite solution.

Table 2 Error terms of solution sets

set	1	2	3	4	5	6	7	8	9	10	mean
ε	0.23	0.12	0.31	0.22	0.39	0.15	0.21	0.25	0.20	0.16	0.22

5 Comparison of different methods

In this section, the performance of the proposed method was evaluated by comparing it with the other four methods. All these methods were applied to the interior layout problem with the same parameter settings stated in Section 4. The experiments were divided into two parts. One was to assess the users' satisfactions and fatigues for different methods with all the 20 users, and the other was to give detailed analyses on different methods with specific users.

5.1 Four comparative methods

The proposed large population algorithm with the similarity-based strategy is expected to have a good performance in exploration on condition of not increasing the user fatigue. In order to verify this, the first comparative method is the same as the proposed method in this study except for adopting a small population, denoted as Method 1.

As mentioned in Section 2, there are three methods of the evolution and decision-making, i.e., *a priori*, *a posteriori* and interactive methods. The proposed method belongs to the interactive methods, and it is necessary to compare it with priori and posteriori methods.

The second and third comparative methods are typical *a posteriori* methods. The second is that the user selects the optimal solution based on his/her preferences from the optimization results obtained by IP-MOEA [10], denoted as Method 2. The third is the same as Method 2 except for adopting the approach in [22] to obtain the optimization results, denoted as Method 3.

The last comparative method is an *a priori* one. In this method, the users precise preferences to different objectives are determined before the evolution, so the bi-objective optimization problem is transformed into a single-objective one, and an evolutionary optimization method is employed to solve it, denoted as Method 4.

5.2 Performance indicators

The following two indicators were employed to compare the user's satisfaction and the time consumption when adopting different methods:

(1) Satisfaction (S metric, for short). The user is asked to score his/her satisfaction to the optimal solution on a 10-point scale, according to both the total cost and the interior layout of a flat. A large value of the S metric indicates that the user is well satisfied.
(2) Time consumption to search for the satisfactory solution (T metric, for short). A small value of the T metric means the method can obtain the satisfactory solution rapidly and ease the user fatigue as a result.

The following three indicators were employed to compare the optimal solutions obtained by different methods:

(1) The mid-point of implicit objective value (MI metric, for short). The larger the value of MI metric, the better the optimal solution.
(2) The mid-point of explicit objective value (ME metric, for short). The smaller the value of ME metric, the better the optimal solution.
(3) Uncertainty (U metric, for short). It refers to the product of the interval widths of all objective values of an optimal solution, the smaller the value of U metric, the better the optimal solution.

The following two indicators were employed to compare the Pareto fronts obtained by different methods:

(1) Best hypervolume (H metric, for short) [28]. Denote the Pareto front obtained by a method as PF^*. Transform an optimization problem into a maximization one by multiplying the objectives to be minimized with -1, the hypervolume calculated with the upper bounds of the interval objective values is called the best hypervolume, and the reference point is [−120000,0]. The larger the value of H metric, the closer PF^* approaches to the true Pareto front.
(2) Imprecision (I metric, for short). It means the average uncertainty of PF^*, where the uncertainty is the product of the interval widths of all objective values. The smaller the value of I metric, the better PF^* is.

5.3 Users' satisfactions and fatigues for different methods

In this subsection, all the 20 users' satisfactions and fatigues for different methods were investigated in blind trial. Before the experiment, all the users were told that the total cost of a flat ranges from about 72000 to 103000, so that they could score a fair S metric for an optimal solution. Each method was successively run in a random order by each user, thus the user did not know which system he/she was running. In the beginning of running the system, the user was asked to input his/her interval preference (shown in Table 1). Since Methods 2, 3, and 4 could adopt precise preferences only, the interval preference would be replaced with its mid-point in the background. Once the user thought that a solution has met his/her requirements or felt too tired to carry on the next evaluation, he/she could stop the system and gave the S metric of the optimal solution. Then the time consumption, i.e. the T metric, would be recorded. The user was required to have a rest at least 20 minutes to eliminate fatigue before running the next system. Table 3 listed the mean values of the S and T metrics obtained from all the users. And the mean values of the MI, ME and U metrics of an optimal solution were also listed to provide more ancillary information.

Table 3 S, T, MI, ME and U metrics obtained by different methods

	S	T	ME	MI	U
Proposed method	8.5	9′15″	81524	85.4	12306
Method 1	8.0	12′36″	82803	84.7	12298
Method 2	8.2	10′52″	81490	82.5	31063
Method 3	7.9	10′32″	82997	82.1	19652
Method 4	7.6	11′02″	83214	83.5	38617

According to Table 3, we can observe that (1) the proposed method achieves the highest mean value of the S metric, suggesting that the users are most satisfied with it. This can also be verified by the mean values of the MI, ME and U metrics of the optimal solutions obtained by the proposed method, which are almost the best among these methods; (2) When the values of the ME metric are larger than 80, most users were either satisfied or tired, and then stopped the system. The time consumption of the proposed method is the smallest, indicating that the proposed method can assist user to search for the satisfactory solution rapidly and alleviate the user fatigue. However, adopting a small population without the similarity-based estimation strategy, Method 1 achieves the worst value of the T metric. In summary, the adaptive precision of the user's interval preference and the similarity-based strategy for estimating interval values of implicit objectives are very effective in searching for the satisfactory solution, and can alleviate the user fatigue.

5.4 Further experiments with specific users

The users' satisfactions and fatigues for different methods were generally assessed in the above subsection. However, for a detailed analysis on the performance of different methods, further experiments with specific users were conducted. In these experiments, users 1 and 2 who have the same interval preferences on different objectives were involved. Their preferences on f_1, i.e., w_1, were both [0.3, 0.7]. For Methods 2, 3, and 4, the values of w_1 were respectively replaced with 0.3, 0.4, 0.5, 0.6, and 0.7 in the background. 20 independent runs of each method were performed, and randomly presented to the users so that any user did not know which method he/she was running. The termination criterion was that the population had evolved for the largest number of generations.

5.4.1 Experimental results and analysis

First, the proposed method was compared with Method 1 to validate the necessity of employing a large population to

Table 4 User's time consumption, number of evolutionary individuals evaluated by user and number of evolutionary individuals searched by proposed method and Method 1 and their p-values

	User's time consumption	# of individuals evaluated by user	# of individuals searched
Proposed method	13′51″	120	2000
Method 1	13′59″	120	120
p-values	0.650	–	–

solve HIMOPs. The population size of Method 1 for optimizing both the explicit and the implicit objectives was the same, 12. The values of its other parameters were the same as those of the proposed method. Table 4 listed the user's time consumption, the number of evolutionary individuals evaluated by the user, and the number of evolutionary individuals searched by an algorithm. Table 5 listed the number of solutions in the Pareto front, the values of both H and I metrics of the Pareto fronts obtained by different methods. In addition, Table 6 listed the values of MI, ME and U metrics of the user's most satisfactory solution obtained by different methods. To determine whether these averages obtained by other methods listed in the three tables have significant differences with those obtained by the proposed method, a commonly used non-parametric test, i.e., Mann-Whitney U test, was carried out on them. The null hypothesis was rejected at a level of significance, α, of 0.05. The corresponding p-values were listed in Tables 4, 5 and 6, where symbol "*" indicates that the proposed method has a significant difference with the other method.

Based on Table 4, we can see that (1) for the two methods, the user evaluates the same number of evolutionary individuals. As a result, they almost have the same user's time consumption, and the corresponding p-value is larger than 0.05. (2) The evolutionary individuals searched by the proposed method are much more than those searched by Method 1, indicating that the proposed method has a better performance in exploration on the premise of the same user fatigue.

Based on Table 5, we can observe that (1) the value of H metric of the Pareto front obtained by the proposed method is significantly larger than that obtained by Method 1. The corresponding p-value is 0, smaller than 0.05, indicates that the proposed method obtains a better Pareto front, which is the result of better performance of the proposed method in exploration. (2) The value of I metric of the Pareto front obtained by the proposed method is also significantly larger than that obtained by Method 1. The possible reason is that the proposed method estimates the values of the implicit objective of a larger number of evolutionary individuals, extending the uncertainty degrees of these estimated evolutionary individuals. (3) The number of solutions in the Pareto-optimal set obtained by the proposed method is significantly more than that obtained by Method 1, suggesting that the proposed method can provide more alternatives for the user than Method 1.

According to Table 6, we can see that (1) the values of both ME and MI metrics of the optimal solution obtained by the proposed method are significantly smaller and larger than those obtained by Method 1, respectively. (2) The value of U metric of the optimal solution obtained by the proposed method has no significant difference from Method 1. Therefore, the optimal solution obtained by the proposed method is better.

It can be concluded that compared with using a small population, employing a large population to solve HIMOPs can generate a better Pareto-optimal set and a more satisfactory solution for the user, which validates the good performance of the proposed method in exploration.

Second, the proposed method was compared with Methods 2 and 3 to investigate the influence of different means of embedding the user's preferences, i.e., interactive and *a posteriori*, on the performance of an algorithm. The parameter settings of Methods 2 and 3 were the same as those of the proposed method, and the results were listed in Tables 2 and 3.

Based on Table 5, we can see that (1) the number of solutions in the Pareto front obtained by the proposed method is smaller than those obtained by Methods 2 and 3, but the difference in the number of the Pareto optimal solutions between the proposed method and Method 2 is not significant, whereas it is significant between the proposed method and Method 3, indicating that interactively providing the user's preferences leads to the user's selection from less and better Pareto optimal solutions at the end of the evolution, alleviating the user's burden. (2) For the same reason, the

Table 5 H and I metrics of Pareto front obtained by different methods and p-values between proposed method and other methods

	# of optimal solutions	p-values	H	p-values	I	p-values
Proposed method	8.9	–	4.4617×10^6	–	12777	–
Method 1	4.2	0.033*	4.1701×10^6	0.000*	7873	0.019*
Method 2	9.6	0.529	4.4553×10^6	0.762	31203	0.001*
Method 3	13.1	0.037*	4.2614×10^6	0.002*	16809	0.070

Table 6 ME, MI and U metrics of optimal solutions obtained by different methods, and p- values between proposed method and other methods

			ME	p-values	MI	p-values	U	p-values
Proposed method			81477	–	91.6	–	12439	–
Method 1			82745	0.049*	89.0	0.031*	9524	0.257
Method 2	w_1	0.3	82496	0.056	89.1	0.041*	30856	0.000*
		0.4	81550	0.151	87.1	0.023*	30109	0.000*
		0.5	81474	0.130	85.7	0.019*	30096	0.000*
		0.6	81447	0.112	85.8	0.025*	32757	0.000*
		0.7	81794	0.257	85.2	0.028*	30066	0.002*
Method 3	w_1	0.3	83738	0.000*	87.7	0.024*	10259	0.046*
		0.4	83039	0.000*	87.2	0.026*	6397	0.040*
		0.5	83051	0.001*	88.3	0.032*	18195	0.032*
		0.6	82805	0.019*	86.4	0.017*	17993	0.035*
		0.7	82607	0.017*	86.4	0.016*	19370	0.030*
Method 4	w_1	0.3	83416	0.000*	89.3	0.044*	33263	0.000*
		0.4	83265	0.001*	88.4	0.030*	27489	0.005*
		0.5	83126	0.003*	87.6	0.014*	36632	0.000*
		0.6	82765	0.021*	86.8	0.011*	44237	0.000*
		0.7	82572	0.025*	86.7	0.011*	39436	0.000*

value of H metric of the Pareto front obtained by the proposed method is larger than those obtained by the other two, but the difference is not significant between the proposed method and Method 2, whereas it is significant between the proposed method and Method 3. (3) The value of I metric of the Pareto front obtained by the proposed method is significantly smaller than that obtained by Method 2, but has no significant difference from Method 3. It can be concluded that the proposed method obtains a better Pareto-optimal set than the other two.

Further, for Methods 2 and 3, five values of w_1, i.e., 0.3, 0.4, 0.5, 0.6 and 0.7, were set. For each value of w_1, the performances of the obtained optimal solutions and their p-values with the proposed method were all listed in Table 6.

According to Table 6, we can see that (1) there is no significant difference between the proposed method and Method 2 for each value of w_1 on the ME metric. The value of the MI metric of the optimal solution obtained by the proposed method is significantly larger than those obtained by Method 2. In addition, the value of the U metric of the

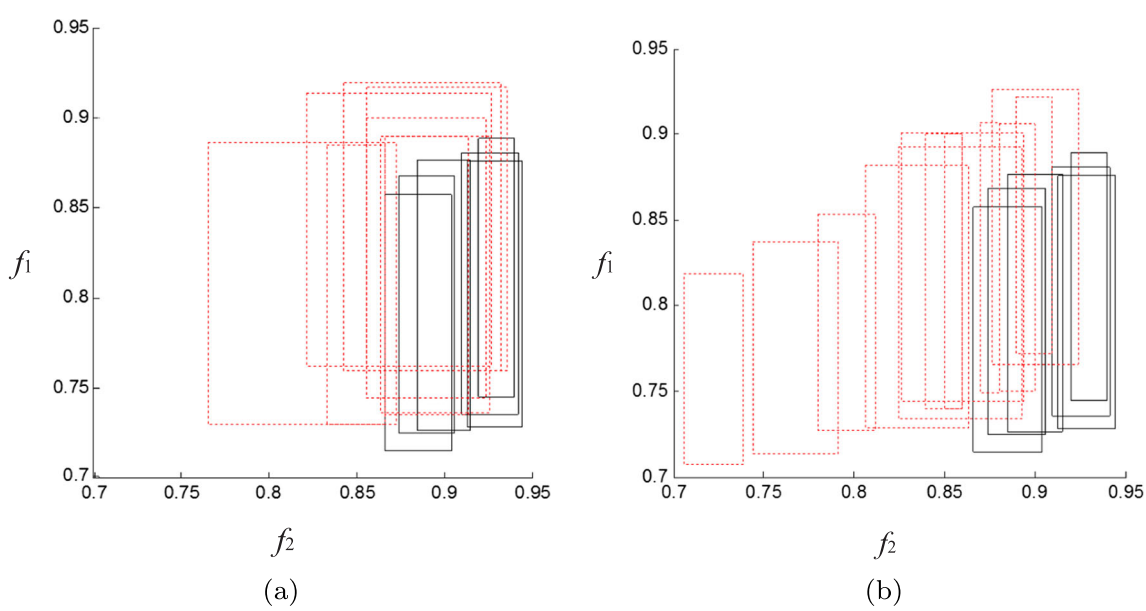


Fig. 7 Pareto Fronts obtained by different methods

Table 7 H and I metrics of Pareto fronts obtained by two different methods in large search space and their p-values

	H	I
Proposed method	4.5186×10^6	13595
Method 2	4.4313×10^6	24187
p-values	0.028*	0.000*

optimal solution obtained by the proposed method is significantly smaller than that obtained by Method 2, indicating that the proposed method is better than Method 2. (2) The values of the ME and MI metrics of the optimal solution obtained by the proposed method are significantly smaller and larger than those obtained by Method 3, respectively. Meanwhile, the value of the U metric of the optimal solution obtained by the proposed method is significantly larger than that obtained by Method 3 when w_1 equals 0.3 and 0.4, whereas, significantly smaller than other cases, implying that the proposed method is better than Method 3. (3) For Methods 2 and 3, the values of the ME and MI metrics of the optimal solution decrease along with the increase of the value of w_1, suggesting that the explicit objective of the most satisfactory solution is better, whereas its implicit objective is worse, which means that none of the two methods can have a good tradeoff between the two objectives. The value of w_1 in the proposed method, however, dynamically changes along with the evolution, leading to a better compromise between the two objectives and the optimal solution with higher quality of the two objectives.

For further performance comparison, Fig. 7 gave the distribution of solutions in the Pareto Fronts obtained by these methods. For clarity, Fig. 7a represented the Pareto Front obtained by the proposed method and Method 2, whereas Fig. 7b meant the Pareto Front obtained by the proposed method and Method 3. The solutions obtained by the proposed method were shown with the solid line rectangles, whereas the solutions obtained by the comparative method were shown with the dashed line rectangles.

From Fig. 7, Method 3 outperforms the other two methods on the distribution of the Pareto Front. The reason is that Method 3 uses the crowd distances as the selection criterion, whereas the proposed method uses the values of the user's preference instead, guiding the solutions to converge toward a preferred local area. In addition, Method 2 uses the hypervolume metric for further comparing solutions, and is slightly better than the proposed method.

Finally, the proposed method was compared with Method 4 to validate the influence of different means to embedding the user's preferences, i.e., interactive and *a priori*, on the performance of an algorithm. For Method 4, five values of w_1, i.e., 0.3, 0.4, 0.5, 0.6 and 0.7, were set. The parameter settings of Method 4 were the same as those of the proposed method. For each value of w_1, the qualities of the obtained optimal solution and their p-values with the proposed method were listed in Table 6.

Based on Table 6, we can observe that (1) the value of the ME/MI metric of the optimal solution obtained by the proposed method is smaller/larger than those obtained by Method 4. At the same time, the value of the U metric of the optimal solution obtained by the proposed method is also smaller than that obtained by Method 4, indicating that the proposed method obtains the optimal solution with higher quality than Method 4. (2) The values of the ME and MI metrics of the optimal solution obtained by Method 4 decrease along with the increase of w_1, suggesting that like Methods 2 and 3, Method 4 cannot have a good tradeoff between the two objectives.

We can conclude from the above analysis that, the proposed method is superior to *a posteriori* and *a priori* methods to different degrees. However, there has no significant difference from Method 2 for some performance indicators. So we will further compare the two methods in a larger search space.

5.4.2 Performance comparison in larger search space

This group of the experiments investigated the capability of the proposed method to explore a larger search space. As mentioned before, the search space of the problem described with formula (14) is rather small. In order to confirm that the proposed method can also be suitable for optimization

Table 8 ME, MI and U metrics of optimal solutions obtained by two different methods in large search space and their p- values

			ME	p-values	MI	p-values	U	p-values
Proposed method			80964	–	91.3	–	13584	–
	w_1	0.3	81178	0.059	89.3	0.022*	19045	0.096
		0.4	81098	0.151	89.0	0.010*	19765	0.082
Method 2		0.5	81069	0.496	88.6	0.011*	21940	0.016*
		0.6	80977	0.820	88.2	0.001*	20390	0.028*
		0.7	80938	0.570	87.6	0.001*	21342	0.016*

problems defined in a larger search space, the search space of the problem described with formula (14) was extended by enlarging the ranges of its decision variables as follows:

$$
\begin{aligned}
&x_1 \in \{4.0, 4.1, 4.2, 4.3, 4.4, 4.5, 4.6, 4.7, 4.8, 4.9, 5.0, 5.1, 5.2\}\\
&x_2 \in \{4.0, 4.3, 4.6, 4.9, 5.2, 5.5, 5.8, 6.1, 6.4, 6.7, 7.0, 7.3\}\\
&x_3 \in \{2.0, 2.1, 2.2, 2.3, 2.4, 2.5, 2.6, 2.7, 2.8, 2.9, 3.0, 3.1, 3.2\}\\
&x_4 \in \{2.0, 2.1, 2.2, 2.3, 2.4, 2.5, 2.6, 2.7, 2.8, 3.1, 3.2, 3.3, 3.4, 3.5, 3.6\}\\
&x_5 \in \{1.0, 1.5, 2.0, 2.5, 3.0, 3.5, 4.0, 4.5, 5.0\}\\
&x_6 \in \{2.6, 2.7, 2.8, 2.9, 3.0, 3.1, 3.2, 3.3, 3.4, 3.5, 3.6, 3.7, 3.8\}\\
&x_7 \in \{1.0, 1.5, 2.0, 2.5, 3.0, 3.5, 4.0, 4.5, 5.0\}
\end{aligned} \tag{17}
$$

As can be observed, in the search space associated with formula (15), the number of alternatives of decision variables $x_1, x_2, x_3, x_4, x_5, x_6$ and x_7 are 13, 12, 13, 17, 9, 13 and 7, respectively. The number of all possible interior layout designs is $13^3 \times 12 \times 17 \times 9 \times 7 = 28235844$, which is far more than that in the problem described with formula (14).

Table 7 listed the values of the H and I metrics of the Pareto front obtained by the proposed method and those obtained by Method 2, and the non-parameter test results about these values. The values of the performance indicators of the optimal solution obtained by the proposed method and those obtained by Method 2 when w_1 is respectively set to 0.3, 0.4, 0.5, 0.6 and 0.7 in the background, and the non-parameter test results about these values were reported in Table 7.

Based on Table 7, we can see that (1) the value of the H metric of the Pareto front obtained by the proposed method is significantly larger than that obtained by Method 2. (2) The value of the I metric of the Pareto front obtained by the proposed method is significantly smaller than that obtained by Method 2, suggesting that the proposed method obtains a better Pareto optimal set than Method 2 in a larger search space.

According to Table 8, we can observe that (1) there is no significant difference between the proposed method and Method 2 for each value of w_1 in the ME metric. The value of the MI metric of the optimal solution obtained by the proposed method is significantly larger than those obtained by Method 2. (2) The value of the U metric of the optimal solution obtained by the proposed method has no significant difference from those obtained by Method 2 when w_1 equals to 0.3, 0.4, but significantly smaller than other cases. Considering all the three performance indicators, the proposed method is better than Method 2.

In summary, we may conclude that it is beneficial to employing a large population, adopting intervals to represent the user's preferences to different objectives, and interactively providing the above preferences when solving HIMOPs, which guarantees that the proposed method can effectively solve HIMOPs.

6 Conclusions

Although HIMOPs are common, there exist few effective methods used to solve them up to now. To effectively solve HIMOPs, we presented an evolutionary optimization algorithm with a user's preferences in this study. In this algorithm, a large population is adopted to improve the performance of the algorithm in exploration. After the population has been divided into several clusters using K-means clustering, a computer is employed to calculate the values of explicit objectives of all evolutionary individuals, and the user only evaluates the values of implicit objectives of the centers of all clusters. For the evolutionary individuals not evaluated by the user, a similarity-based strategy is presented to estimate the interval values of their implicit objectives so as to alleviate user fatigue. To reflect the uncertainty of the user's preferences embedded in the course of the evolution, the user's preferences to different objectives are expressed with intervals, and made precise by solving another optimization problem. After adopting an interval dominance relation to sort different evolutionary individuals, the user's satisfactory degrees to them based on the above preference are calculated to further distinguish evolutionary individuals with the same rank. As a consequence, the search of the algorithm is guided toward the user's preferred region.

We applied the proposed algorithm to the interior layout problem, a typical optimization one with both interval parameters in the explicit objective and interval value of the implicit objective, and compared it with four other optimization algorithms. The experimental results showed that the proposed method outperforms the others in almost all aspects when solving HIMOPs.

It is worth noting that the user's preferences to objectives were represented as intervals in this study. In addition, the uncertainties of the optimization problem considered here either resulted from the parameters contained in the objectives whose values are intervals, or the evaluation results of the user on implicit objectives expressed with intervals. In fact, the user's preferences or the uncertainties of an optimization problem have other characters, e.g., stochastic or fuzzy. At this point, how to solve these optimization problems needs to be further studied. In addition, when $P + Q > 3$, HIMOP described in formula (1) is also a many-objective optimization problem. In this situation, it is considerably difficult to obtain the domination relationships among solutions, because of not only the large number of objectives, but also their uncertain objective values. During the evolution, a user's preference plays a much more

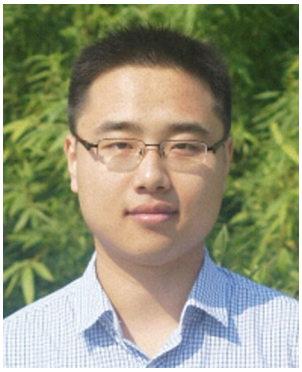
important role in selecting superior solutions than before. Hence, investigating the behavior of the proposed method on hybrid interval many-objective optimization problems is also our future work.

Acknowledgments This work was jointly supported by National Natural Science Foundation of China with grant No. 61375067 and 61403155, National Basic Research Program of China (973 Program) with grant No. 2014CB046306-2, and Natural Science Foundation of Jiangsu Province with grant No. BK2012566. Thanks to Dr. Jianyong Sun for linguistic advice.

References

1. Funes P, Bonabeau E, Herve J, Morieux Y (2004) Interactive multi-participant tour allocation. In: Proceedings of IEEE congress on evolutionary computation, pp 1699–1705
2. Zhou Y, Gong DW, Zhang Y (2007) Evolutionary optimization methods for hybrid index optimization problem and application. Control Decis 22:352–356. doi:10.3321/j.issn:1001-0920.2007.03.025
3. Cheng ZQ, Dai LK, Sun YX (2004) Feasibility analysis for optimization of uncertain systems with interval parameters. Acta Autom Sin 30:455–459
4. Gong DW, Qin NN, Sun XY (2011) Evolutionary algorithms for optimization problems with uncertainties and hybrid indices. Inf Sci 181:4124–4138. doi:10.1016/j.ins.2011.05.011
5. Branke J, Deb K, Miettinen K, Slowinski R (2008) Multiobjective optimization - interactive and evolutionary approaches. Lecture Notes in Computer Science. Springer, Heidelberg, p 5252
6. Deb K, Pratap A, Agarwal S, Meyarivan T (2002) A fast and elitist multiobjective genetic algorithm: NSGA-II. IEEE Trans Evol Comput 6:182–197. doi:10.1109/4235.996017
7. Kamalian R, Takagi H, Agogino MA (2004) Optimized design of MEMS by evolutionary multi-objective optimization with interactive evolutionary computation. In: Proceedings of genetic and evolutionary computation conference, pp 1030–1041
8. Brintrup AM, Ramsden J, Tiwari A (2005) Integrated qualitativeness in design by multi-objective optimization and interactive evolutionary computation. In: Proceedings of IEEE congress on evolutionary computation, pp 2154–2160
9. Brintrup AM, Ramsden J, Tiwari A (2007) An interactive genetic algorithm-based framework for handling qualitative criteria in design optimization. Comput Ind 58:279–291. doi:10.1016/j.compind.2006.06.004
10. Limbourg P, Aponte DES (2005) An optimization algorithm for imprecise multi-objective problem functions. In: Proceedings of IEEE congress on evolutionary computation, pp 459–466
11. Li FY, Luo Z, Rong JH, Zhang N (2013) Interval multi-objective optimisation of structures using adaptive Kriging approximations. Comput Struct 119:68–84. doi:10.1016/j.compstruc.2012.12.028
12. Coello CAC (2000) Handling preferences in evolutionary multiobjective optimization: A survey. In: Proceeding of congress on evolutionary computation, pp 30–37
13. Parmee IC, Cvetkovič D, Watson AH, Bonham CR (2000) Multiobjective satisfaction within an interactive evolutionary design environment. Evol Comput 8:197–222. doi:10.1162/106365600568176
14. Chaudhuri S, Deb K (2010) An interactive evolutionary multiobjective optimization and decision making procedure. Appl Soft Comput 10:495–511. doi:10.1016/j.asoc.2009.08.019
15. Deb K, Sinha A, Korhonen PJ, Wallenius J (2010) An interactive evolutionary multi-objective optimization method based on progressively approximated value functions. IEEE Trans Evol Comput 14:723–739. doi:10.1109/TEVC.2010.2064323
16. Sinha A, Korhonen P, Wallenius J, Deb K (2010) An interactive evolutionary multi-objective optimization method based on polyhedral cones. In: Proceedings of learning and intelligent optimization conference, pp 318–332
17. Horn J, Nafpliotis N, Goldberg DE (1994) A niched Pareto genetic algorithm for multiobjective optimization. In: Proceedings of the 1st IEEE congress on evolutionary computation, pp 82–87
18. Deb K, Kumar A (2007) Light beam search based multiobjective optimization using evolutionary algorithms. In: Proceedings of IEEE Congress on Evolutionary Computation, pp 2125–2132
19. Greenwood GW, Hu X, D'Ambrosio JG (1996) Fitness functions for multiple objective optimization problems: combining preferences with Pareto rankings. In: Proceedings of 4th workshop on foundations of genetic algorithms, pp 437C455
20. Bentley PJ, Wakefeild JP (1998) Finding acceptable solutions in the Pareto-optimal range using multiobjective genetic algorithms. Soft computing in engineering design and manufacturing. Springer, London, pp 231–240
21. Wang ZJ, Li KW, Wang WZ (2009) An approach to multiattribute decision making with interval-valued intuitionistic fuzzy assessments and incomplete weights. Inf Sci 179:3026–3040. doi:10.1016/j.ins.2009.05.001
22. Gong DW, Qin NN, Sun XY (2010) Evolutionary algorithms for multi-objective optimization problems with interval parameters. In: Proceedings of 5th IEEE international conference on bio-inspired computing: theories and applications, pp 411–420
23. Gong DW, Yuan J, Ma XP (2008) Interactive genetic algorithms with large population size. In: Proceedings of IEEE congress on evolutionary computation, pp 1678–1685
24. Gong DW, Qin NN, Sun XY (2010) Novel evolutionary optimization algorithm for problems with hybrid indices. Control Theory Appl 27:769–774
25. Wilpon JG, Rabiner L (1985) A modified K-means clustering algorithm for use in isolated work recognition. IEEE Transactions on Acoustics. Speech, and Signal Processing 33:587–594. doi:10.1109/TASSP.1985.116458
26. Ishibuchi H, Tanaka H (1990) Multiobjective programming in optimization of the interval objective function. Eur J Oper Res 48:219–225. doi:10.1016/0377-2217(90)90375-L
27. Xu GL, Liu YJ (2008) A new ranking approach for interval numbers in uncertain multiple-attribute decision making problems. Stat Decis 19:154–157
28. Zitler E, Thiele L (1998) Multiobjective optimization using evolutionary algorithms - a comparative case study. In: Proceedings of 5th international conference on parallel problem solving from nature, pp 292–301

Dunwei Gong was born in Xuzhou in 1970. He received the B.S. degree from China University of Mining and Technology, Xuzhou, P.R. China, in 1992, the M.S. degree from Beijing University of Aeronautics and Astronautics, Beijing, P.R. China, in 1995, and the Ph.D. from China University of Mining and Technology in 1999. Since 2004, he has been a Professor of China University of Mining and Technology. Up to date, he has co-authored six monographs and more than 100 journal papers. His current research interests are evolutionary computation and its applications.



Yiping Liu was born in Xuzhou in 1990. He received the B.S. degree in Electrical Engineering and Automation from China University of Mining and Technology, Xuzhou, P.R. China, in 2012. He is working for Ph.D. degree in the School of Information and Electronic Engineering, China University of Mining and Technology. His main research interests are evolutionary computation and multi-objective optimization.

Xinfang Ji was born in Taizhou in 1987. She received the B.S. degree in Electrical Engineering and Automation in 2010, and the M.S. degree in control theory and control engineering in 2013 from China University of Mining and Technology, Xuzhou, P.R. China. Since 2013, she has been a lecturer in Department of Mechanical, Power and Information Engineering, China University of Mining and Technology, Yinchuan College. Her main research interest is interactive evolutionary computation.

Jing Sun was born in Lianyungang in 1975. She received the B.S. degree in mathematics education from Center China Normal University, Wuhan, P.R. China, in 1997, the M.S. degree in applied mathematics from China University of Mining and Technology, Xuzhou, P.R. China, in 2003, and the Ph.D. in control theory and control engineering from China University of Mining and Technology, Xuzhou, P.R. China, in 2012. Since 2010, she has been an associate professor in College of science, Huai Hai Institute of Technology. Her main research interests are evolutionary computation and interval optimization.